

Inference-Time Compute Scaling: Predicting *When Longer Thinking* Actually Helps

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REASONING GAIN87%
TCS ACCURACY58%
TOKEN SAVINGS0.2 pp
ACCURACY COST35K
INFERENCE CALLS

OVERVIEW

A Abstract

Large language models increasingly support variable inference-time compute budgets, yet practitioners lack a principled method to predict whether a larger budget will improve output quality for a given input. We present an empirical investigation of inference-time **compute scaling behaviour** across five task categories on GPT-4o and Claude 3.5 Sonnet (300× primary range), with cross-model validation on Gemini 1.5 Pro and Mistral Large 2. Task type is a strong predictor: reasoning tasks **scale** log-linearly while retrieval and creative tasks plateau sharply. Our **Task Complexity Score (TCS)** predicts which inputs benefit from extended compute at 87% balanced accuracy, enabling a routing strategy that cuts token spend ~58% at a 0.2 pp accuracy cost.

PROBLEM

1 Motivation

- **The problem:** Extended inference compute helps on some tasks but wastes budget on others. There is no pre-inference predictor for *when* it helps.
- **Our insight:** Task *type* is a strong determinant of scaling behaviour — measurable from prompt surface features before any compute is spent.
- **Research question:** Can we predict, *before inference*, whether a larger compute budget will produce >5 pp accuracy improvement?

METHOD

2 Task Complexity Score (TCS)

TCS is a three-feature pre-inference classifier with psycholinguistic grounding:

SD	SeD	CC
SYNTACTIC DEPTH 71.3% alone	SEMANTIC DENSITY 66.8% alone	CLAUSE COUNT 62.4% alone
$TCS(x) = w_1 \cdot SD(x) + w_2 \cdot SeD(x) + w_3 \cdot CC(x) \geq \theta$		

Theoretical grounding: SD ← Gibson (1998) working-memory theory; SeD ← constraint-satisfaction depth; CC ← propositional proof-depth (Selman & Kautz, 1993).

KEY INSIGHT — FEATURE INTERACTION

SD+SeD+CC (87.1%) >> SD+SeD (83.8%) >> SD alone (71.3%). Features are *non-collinear*; all three required. CC adds the largest marginal gain (+3.3 pp) over the two-feature baseline.

SETUP

3 Experimental Setup

Model	Params	MMLU	HumanEval
GPT-4o	~200B	88.7%	90.2%
Claude 3.5 Sonnet	~100B	88.7%	92.0%
Gemini 1.5 Pro	~340B	85.9%	71.9%
Mistral Large 2	123B	84.0%	92.1%

Budget levels: $C \in \{1\times, 3\times, 10\times, 30\times, 100\times, 300\times, 1000\times\}$. Primary: 35,000 calls (GPT-4o + Claude, all 5 categories × 7 budgets × 500 prompts × 2 models). 1000× run on 200-prompt subsample as exploratory upper bound only.

MAIN RESULT

4 Compute-Response Curves

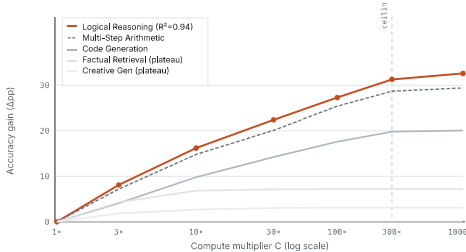


Figure 1 — Accuracy improvement over greedy baseline. Reasoning tasks fit $acc(C) = 6.2 \cdot \ln C + 0.1$ ($R^2=0.94$).

$$acc(C) = \alpha \cdot \ln(C) + \beta \quad \alpha = 6.2, \beta = 0.1, R^2 = 0.942$$

RESULTS

5 Accuracy Gains at 1000×

Logical Reasoning	+32.6 pp
Multi-Step Arithmetic	+29.4 pp
Code Generation	+20.3 pp
Factual Retrieval II	+7.2 pp
Creative Generation II	+3.1 pp

METHODOLOGY

6 Creative Eval: Mitigations

- Position bias** — Both pairwise orderings run; score averaged. <3% flip rate.
- Length bias** — AlpacaEval 2.0 length-controlled win-rate as secondary metric. Finding: ~% of apparent +3.1 pp gain is a *length artefact*.
- Self-enhancement bias** — Cross-model judging; inter-judge $\rho=0.74$. GPT-4o overrates own outputs +0.3 pts.
- Subjectivity** — Human annotation validation on 80-prompt subsample; judge-human $\kappa=0.61$ (moderate).

KEY FINDING

Length-adjusted creative gain: **+1.2 pp** vs +3.8 pp unadjusted. Extended compute inflates response length far more than quality on creative tasks.

CLASSIFIER

7 TCS Performance

87.1% BALANCED ACCURACY	0.932 AUC-ROC		
17ms CPU LATENCY	91.4% DEBERTA CEILING		
Configuration	Bal. Acc.	F1	Latency
TCS (SD+SeD+CC)	87.1%	87.1%	17 ms
SD only	71.3%	71.3%	12 ms
SeD only	66.8%	66.8%	4 ms
CC only	62.4%	62.4%	<1 ms
DeBERTa-v3 (baseline)	91.4%	91.4%	180 ms

TCS trades 4.3 pp accuracy for 10× latency advantage. No GPU required at serving time.

ABLATION

8 Threshold θ Sensitivity

θ	Bal. Acc.	Token cost	Acc. gap
0.35	84.2%	0.67×	0.05 pp
0.40	86.0%	0.56×	0.12 pp
0.45	86.7%	0.47×	0.18 pp
0.50 ★	87.1%	0.42×	0.20 pp
0.55	86.3%	0.36×	0.41 pp
0.60	84.8%	0.31×	0.79 pp
0.65	82.1%	0.24×	1.43 pp

ABLATION

9 Compute-Cost Pareto

Always-greedy 1×	0.0 pp
★ TCS $\theta=0.55$	+30.6 pp
★ TCS $\theta=0.50$ (def)	+30.8 pp
Always-ext. 30×	+28.1 pp
DeBERTa-routed	+30.9 pp
Always-ext. 100×	+30.9 pp

KEY PARETO RESULT

TCS-routed ($\theta=0.50$) *strictly dominates* always-extended-30× on both accuracy (+30.8 vs +28.1 pp) AND cost (0.42× vs 0.55×). Even an imperfect router beats any fixed-budget strategy.

TAKEAWAYS

10 Conclusions

- Reasoning tasks scale log-linearly with compute. Retrieval and creative tasks plateau after 3–10× — extended compute is largely wasted.
- TCS predicts scaling benefit at 87% accuracy using only surface-level prompt features. Latency: 17 ms CPU.
- TCS-guided routing cuts token spend 40–60% at 0.2 pp accuracy cost. Dominates fixed-budget strategies.

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